

2020

Short Q.

Q. Discuss the importance of Sample ACF & PACF in time series Modeling.

In time series Modeling Building ACF & PACF are crucial for identifying the structure of the data and selecting an appropriate model. They help diagnose the relationships b/w obs over time.

ACF It measures the correlation between time series and its lagged values. (How much a time point is related to its past points)

PACF PACF measures the correlation between a time series and its lagged values, after removing the effects of shorter lags. PACF helps isolate the direct relationship b/w the series & each specific lag

Importance

=> ACF identifies how past values (lags) influence the current value, useful to identify the No of lagged Forecast Errors (2) included in MA Model. (or help choosing MA terms).

=> PACF isolates the direct relationship b/w specific lags and the current value help choosing AR(p) terms

- ⇒ ACF reveals seasonality if there are repeating spikes at regular intervals
- ⇒ PACF shows which lags have the most impact and avoids including unnecessary lags, preventing overfitting.
- ⇒ ACF also helps check stationarity, A slow decay suggests non-stationarity.
- ⇒ Together ACF & PACF help select the right number of AR and MA terms for models like ARIMA, in time series.

b. Explain meaning and Importance of updating forecasts.

Updating Forecasts

It refers to the process of revising predictions based on new data or past forecast errors. As new information becomes available, forecast models can be recalibrated to improve their accuracy. This ensures that the predictions remain relevant and reflect the latest trends or changes in the data.

Importance:

1. Improved Accuracy

regularly updating forecasting

based on new data or past errors, helps reduce the gap b/w the predicted and actual outcomes, making future predictions more reliable.

2. Adapting to change:

Things change over time (like markets or trends, environments etc) updating forecasts allows models to adapt to these changes & maintain their relevance over time, (helps to models stay current).

3. Error Correction:

By analyzing past errors, we can adjust our current model to prevent repeating the same mistakes & improving future predictions.

4. Real Time Decision making or Faster Decisions:

In areas like finance or weather, updated forecasts, give more accurate information, helping people make smarter decisions quickly.

5. Tracks changes:

If your model starts to give worse predictions, updating it helps catch and fix the problems.

Hence updating forecasts ensures models remain accurate, adaptable and useful for decision making in dynamic environments.

C. Describe the procedure of finding the iterative least square estimates of MA(1) process given by:

$$X_t = \mu + \theta Z_{t-1} + Z_t$$

$$X_t = \mu + Z_t + \theta Z_{t-1}$$

$$Z_t = X_t - \mu - \theta Z_{t-1}$$

Sum of squared errors.

$$S = \sum_{t=1}^n Z_t^2 = \sum_{t=1}^n (X_t - \mu - \theta Z_{t-1})^2$$

The procedure to find the iterative least square estimates of parameters, follows these steps.

1. Initialize the Parameters:-

Start with initial guess for the parameters.

$(\mu)^0$ (Initial guess for the mean)

$(\theta)^0$ (Initial guess for the MA parameter)

2. Compute residuals:-

calculate the residuals using the current estimates for parameters μ^i & θ^i .

$$Z_t^i = X_t - \mu^i - \theta^i Z_{t-1}^i$$

where r_1 is the autocorrelation at lag 1.

Estimation of Z_0 :

As $\{Z_t\}$ is a purely random process therefore it is a stationary process. Hence we can use

$$Z_0 = E[Z_t]$$

$$Z_0 = 0$$

Assuming $\{Z_t\}$ is a zero mean process

Least sq estimation of MA(1) process

Iteration 1, 0: considers MA(1) process

$$y_t = \mu + Z_t + \theta(Z_{t-1})$$
$$\mu = \mu - \theta(Z_{t-1})$$

To obtain LSE of MA parameter, objective function can be written as

$$S = \sum_{t=1}^n z_t^2 = \sum_{t=1}^n (y_t - \mu - \theta \cdot z_{t-1})^2$$

 we can't perform analytical optimization due to presence of z_{t-1} in objective function. One can perform numerical optimization by following algorithm

Iteration - 0:

$$z_1^{(0)} = y_1 - \mu^{(0)} - \theta^{(0)} z_0$$

$$z_2^{(0)} = y_2 - \mu^{(0)} - \theta^{(0)} z_1^{(0)}$$

$$\vdots$$

$$z_n^{(0)} = y_n - \mu^{(0)} - \theta^{(0)} z_{n-1}^{(0)}$$

we calculate

$$S^{(0)} = \sum_{t=1}^n (z_t^{(0)})^2$$

Iteration - 1:

update estimates of μ & θ get $\mu^{(1)}$ and $\theta^{(1)}$, similarly

$$z_1^{(1)} = y_1 - \mu^{(1)} - \theta^{(1)} z_0$$

$$z_2^{(1)} = y_2 - \mu^{(1)} - \theta^{(1)} z_1^{(1)}$$

$$\vdots$$

$$z_n^{(1)} = y_n - \mu^{(1)} - \theta^{(1)} z_{n-1}^{(1)}$$

thus we calculate

$$S^{(1)} = \sum_{t=1}^n (z_t^{(1)})^2$$

This process continues till S converges to a minimum. The value of μ and θ at the minimum level of S are taken as LSEs.

d. If $\{X_t\}$ follows an AR(2) process with zero mean, then find the Yule-Walker estimates of the parameters of the Model.

AR(2) process is: with zero mean:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

As we know for an AR(P) process

$$e_k = \phi_1 e_{k-1} + \phi_2 e_{k-2} + \dots + \phi_p e_{k-p}$$

Using Theory of Method of moments Estimation:

$$\hat{\pi}_k = \hat{\phi}_1 \hat{\pi}_{k-1} + \hat{\phi}_2 \hat{\pi}_{k-2} + \dots + \hat{\phi}_p \hat{\pi}_{k-p}$$

for AR(p),

$$e_k = \phi_1 e_{k-1} + \phi_2 e_{k-2} + \dots + \phi_p e_{k-p}$$

for AR(2):

$$e_k = \phi_1 e_{k-1} + \phi_2 e_{k-2}$$

$$x_k = \phi_1 x_{k-1} + \phi_2 x_{k-2}$$

at $k=1$:

$$x_1 = \hat{\phi}_1 + \hat{\phi}_2 x_1 \quad \text{--- (i)}$$

at $k=2$,

$$x_2 = \hat{\phi}_1 x_1 + \hat{\phi}_2 \quad \text{--- (ii)}$$

in matrix form, using (i) & (ii)

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ x_1 & 1 \end{bmatrix} \begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ x_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \frac{1}{1 - x_1^2} \begin{bmatrix} 1 & +x_1 \\ -x_1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} (x_1 - x_1 x_2) / (1 - x_1^2) \\ (-x_1^2 + x_2) / (1 - x_1^2) \end{bmatrix}$$

required Yule
Walker estimates;

2021

Short Q,

(a) Describe the Model building stages suggested by Box and Jenkins.

Repeated

b. what is meant by Residual Analysis? Describe its role in model diagnostic checking.

Residual Analysis is a technique used to evaluate how well a statistical model fits the data. Residuals analysis helps to determine if the model is appropriately capturing the patterns of in the data. it's a diagnostic tool for checking if the assumptions of the model are met.

Residuals represents the Error or deviation for each data point, calculated as

$$\text{Residual} = \text{obs value} - \text{predicted value}$$

Role Model in Diagnostic checking:

=> Check for Bias

Ideally, residuals should be randomly distributed around zero. Patterns or trends in residuals suggest that the model is missing important information or relationships.

⇒ Normality.

in many models, like in linear regression models, residuals are expected to follow a normal distribution. Deviations from normality may indicate problem with model.

⇒ Homoscedasticity:

The spread of residuals should be consistent across predicted values. If residuals have increasing or decreasing variance (heteroscedasticity) it may suggest the model is inappropriate.

⇒ Independence:

Residuals should be indep of each other. If residuals show autocorrelation, the model may not be properly accounting for temporal patterns.

That's how residuals analysis helps ensure the model's assumptions are valid, and it provides insights into potential improvements for model accuracy.

c. what is portmanteau test? Describe its role in model diagnostic checking.
Repeated.

d. Describe the Forecast Errors and its role in updating the forecasts.

Forecast Errors refers to the diff b/w the actual value of a var & the forecasted value of that var. They are crucial in assessing the accuracy of a forecasting model.
Forecast Error = obs value - Forecasted value.

Types

1. Mean Absolute Error MAE -

it measures the Average Absolute distance b/w each obs & the mean of all observations. $\Rightarrow \frac{1}{n} \sum_{i=1}^n |O_i - F_i|$

2. MSE

Calculates the average of squared differences b/w obs & predicted values. $\Rightarrow \frac{1}{n} \sum_{i=1}^n (O_i - F_i)^2$

3. Root Mean square Error (RMSE):-

The square root of the mean squared Error, bringing it back to the original units. $\Rightarrow \sqrt{\frac{1}{n} \sum_{i=1}^n (O_i - F_i)^2}$

4. Mean Absolute Percentage Error $\Rightarrow \frac{1}{n} \sum_{i=1}^n \left| \frac{O_i - \text{Forecasted}_i}{\text{Observed}_i} \right| \times 100$

The average of the absolute Percentage Errors, Expresses errors as percentages of actual values, making it easy to compare across diff data sets.

Rule in updating Forecasts:

1. Model calibration:

Forecast errors highlight where a model may be systematically underperforming. By analyzing these errors, you can adjust model parameters to better fit historical data.

2. Model selection:

Comparing errors across different forecast models helps in selecting the best model. Models with lower forecast errors are typically preferred.

3. Adaptive learning:

Errors are used in adaptive learning systems to continuously update and refine the model, incorporating new data and correcting previous inaccuracies.

4. Error Analysis and model Improvement.

By analyzing Forecast errors, patterns or biases, can be identified leading to model improvements and the inclusion of additional relevant features.

5. Benchmarking.

Forecast Errors help set standards for acceptable accuracy level and allow for comparative analysis against industry benchmarks or historical performance.

6. Risk management & Decision Support.

Forecast errors help quantify uncertainty, enabling better risk management and more informed decision-making by preparing for potential deviations from predictions.